ChatClinic Segmentation Tools

Five segmentation plugins for the ChatClinic

Topic: Segmentation

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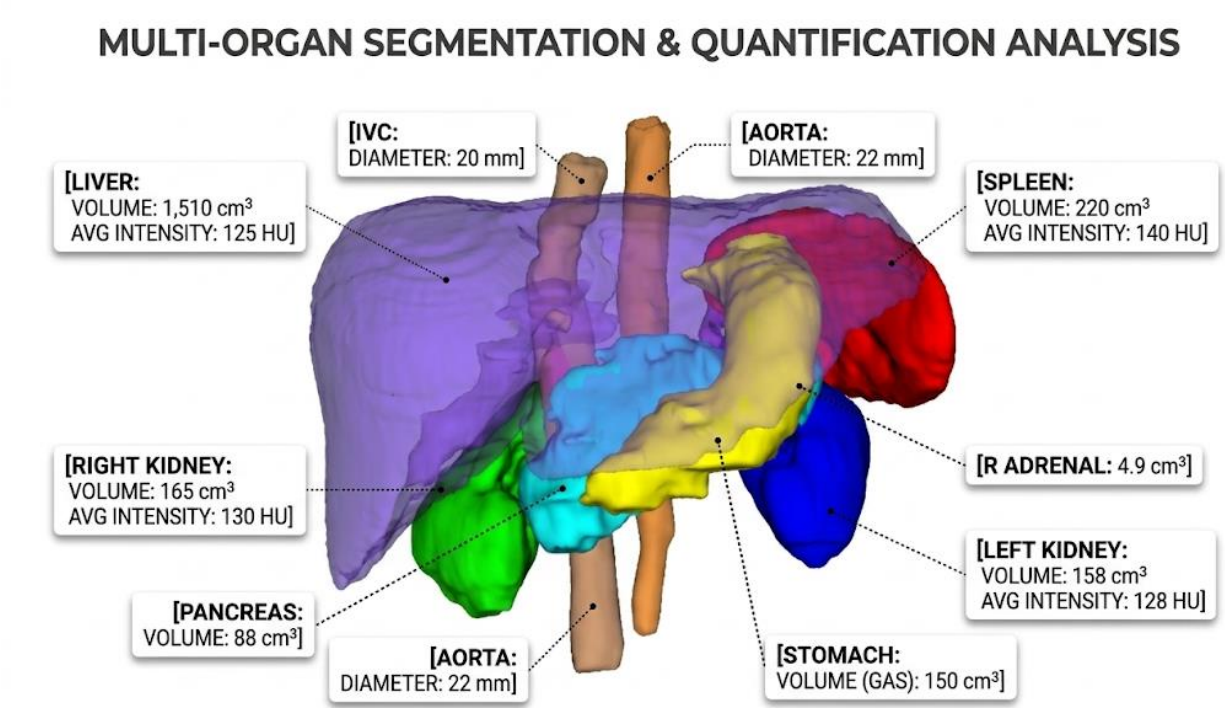
KAIST AI619 — AI for Medical Imaging and Signals · Spring 2026

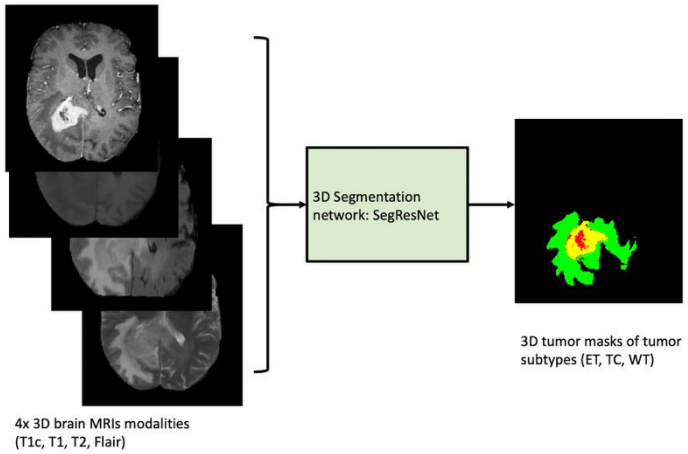
Why Segmentation Tool Should Exist?

- <u>No lesion/organ extraction capability exists in ChatClinic</u>
- <u>Segmentation is a prerequisite for subsequent analyses</u> (e.g., volumetric measurement, lesion quantification, and treatment response evaluation)

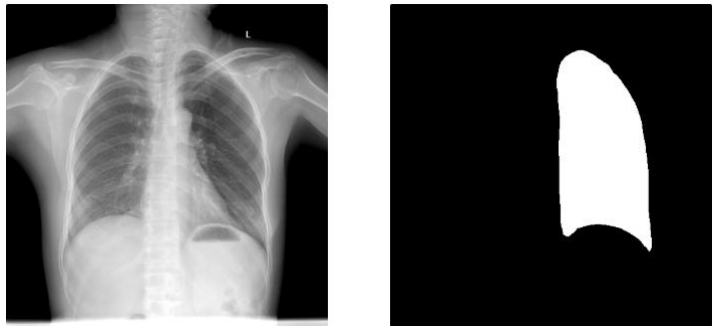
→ Introduce segmentation tools that can be invoked **<u>on-demand</u>** to the extent the user requires



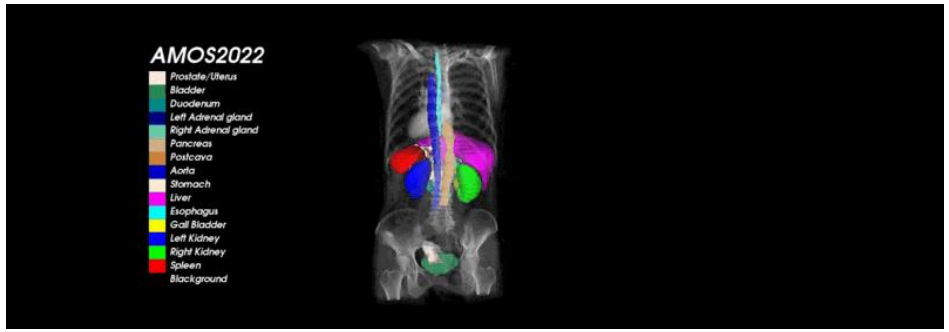
Segmentation Tools — Overview



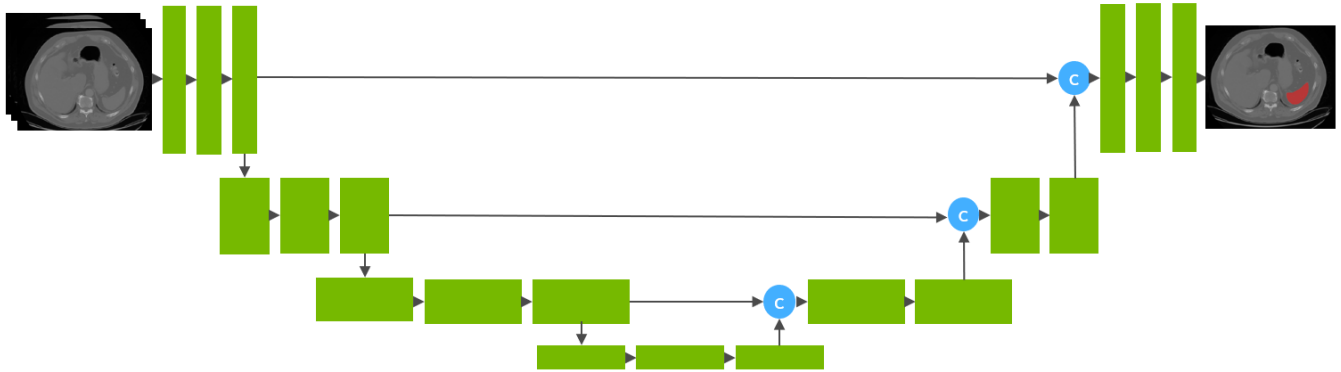
1. Brain Tumor (@seg_brain)



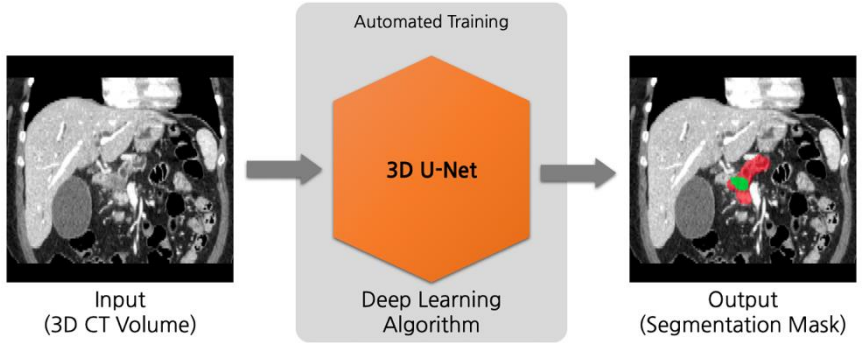
2. Lung (@seg_lung)



3. Multi-Organ (@seg_organ)

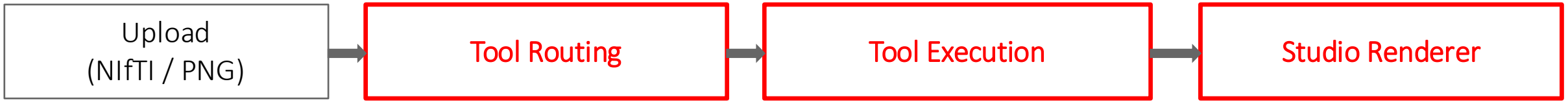


4. Spleen (@seg_spleen)



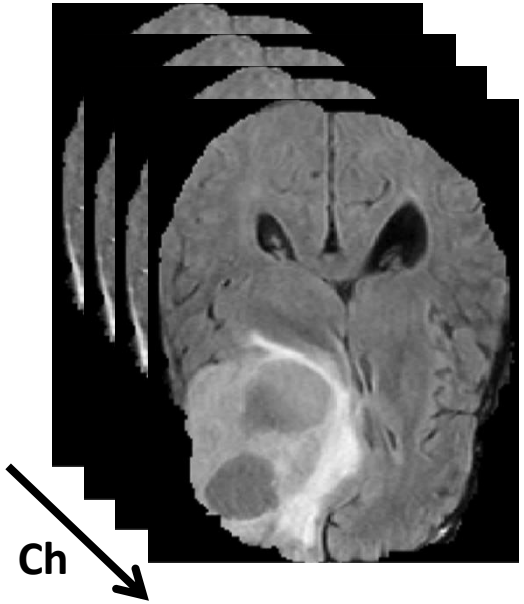
5. Pancreas Tumor (@seg_pancreas)

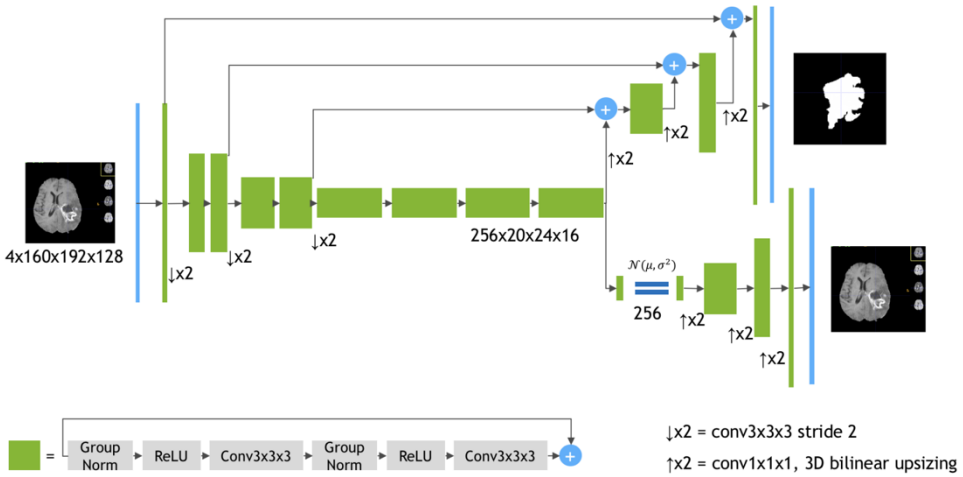
Segmentation Tools — Overview

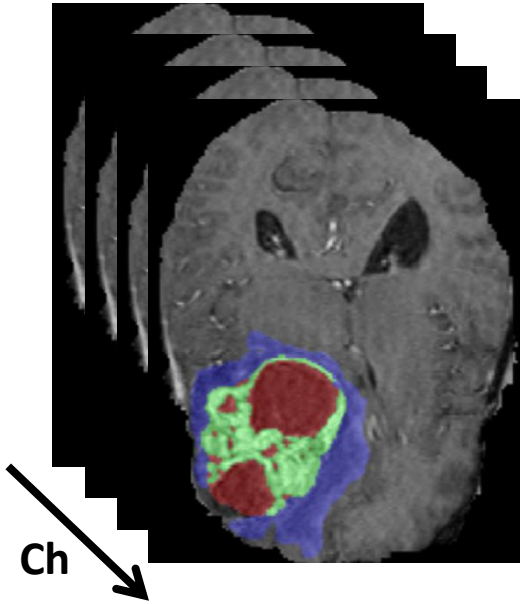


Tool	Modality / Target	Model	Weights
`@seg_brain`	3D MRI · Brain tumor (ET/TC/WT)	SegResNet [1]	MONAI bundle [9]
`@seg_lung`	2D CXR · Left / Right Lung	PSPNet [2]	TorchXRayVision [4]
`@seg_organ`	3D CT · 15 abdominal organs	MaskSAM [5]	Trained by us
`@seg_spleen`	3D CT · Spleen	Residual U-Net [7]	MONAI bundle [9]
`@seg_pancreas`	3D CT · Pancreas + Tumor	DiNTS [8]	MONAI bundle [9]

@seg_brain — Brain Tumor Segmentation







Input

- 3D MRI volume (NIfTI)
- 4 channels: T1, T1ce, T2, FLAIR
- Input Size: 224 x 224 x 224

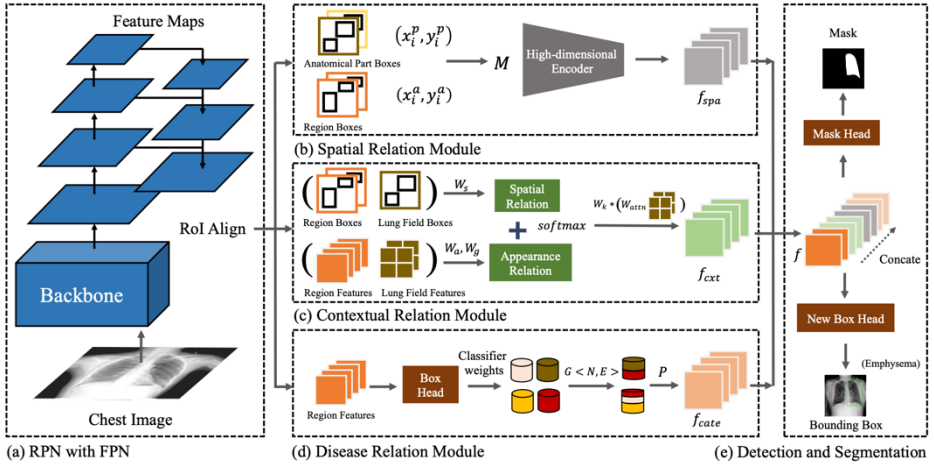
Model

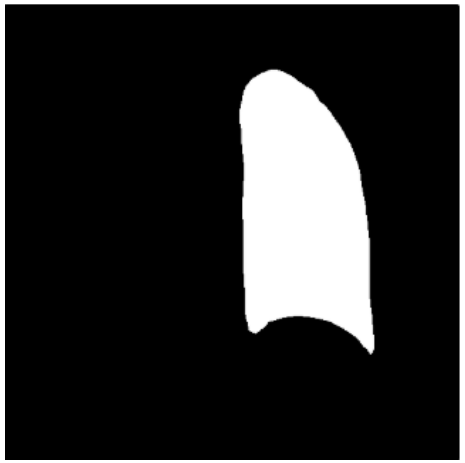
- SegResNet [1]
- MONAI bundle: BraTS mri segmentation [9]

Output

- 3-class mask
- ET / TC / WT tumor subregions
  - ET: Enhancing Tumor
  - TC: Tumor Core
  - WT: Whole Tumor

@seg_lung — Chest X-ray Lung Segmentation





Input

- 2D chest X-ray (PNG / JPG)
- Input Size: 320 x 320

Model

- PSPNet [2]
  (from SARNet [3], via TorchXRayVision [4])
- 14-class: lungs, heart, aorta, spine, …

Output

- Binary lung mask

post-processing: 14-class output → channels [4, 5] (L / R Lung) → binary lung mask

@seg_organ — MaskSAM: Auto-prompt SAM [5]

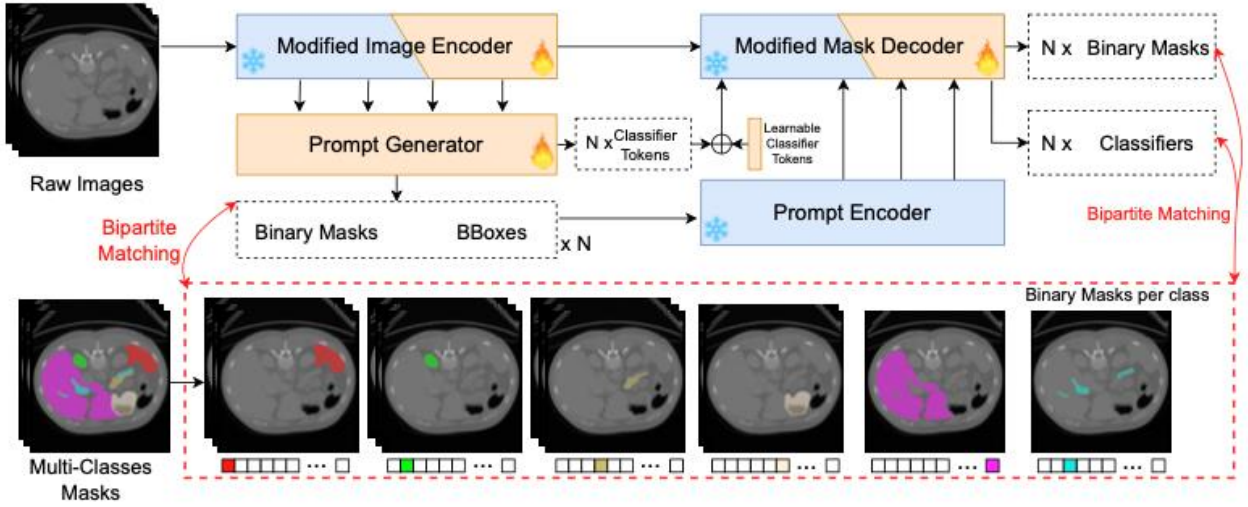


Figure 2. The overview architecture of our proposed MaskSAM.

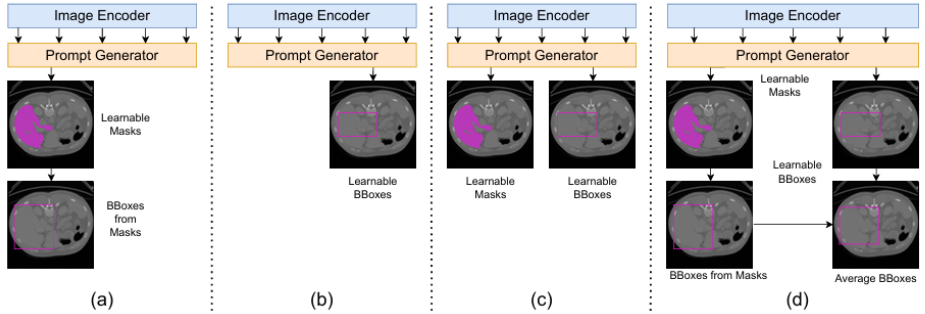
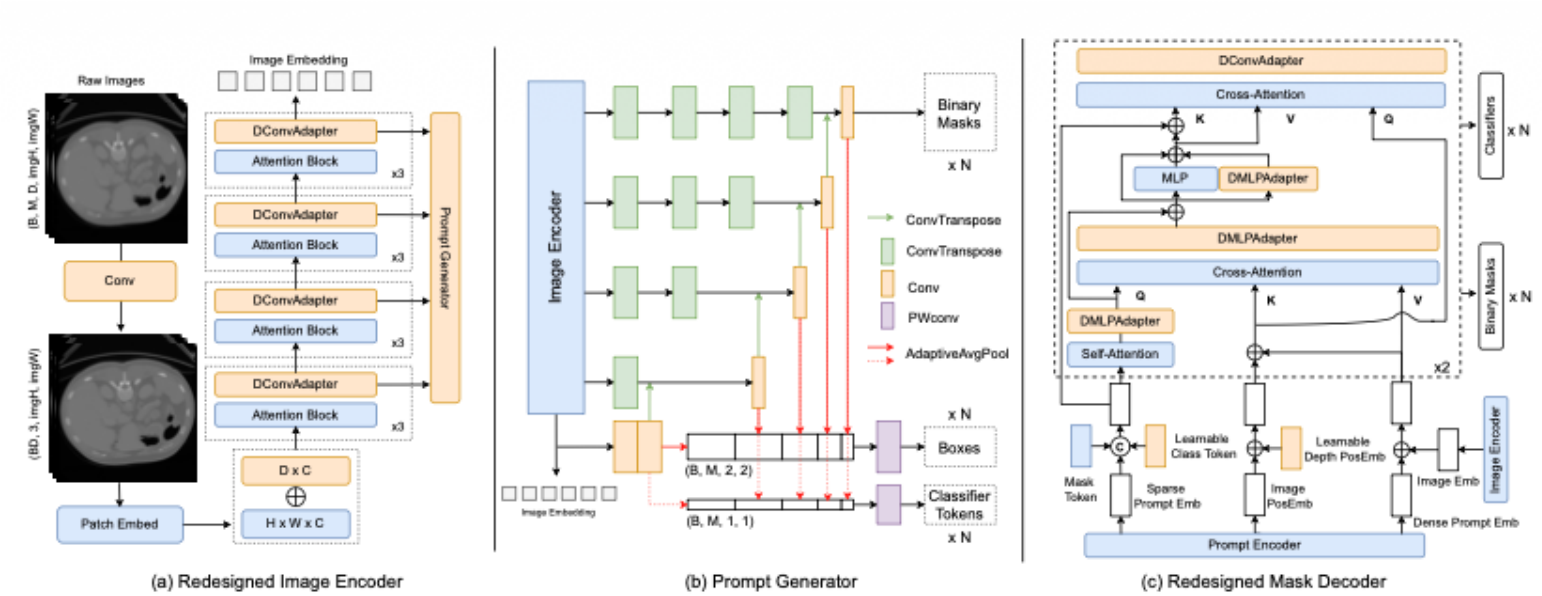


Figure 5. (a) A prompt generator with learnable masks. (b) A prompt generator with learnable boxes. (c) A prompt generator with learnable masks and learnable boxes. (d) A prompt generator with learnable masks and average boxes.

Contributions

- A novel mask classification **prompt-free SAM** framework, MaskSAM
- **Prompt generator** generates auxiliary masks and bound boxes as prompts to solve the requirements of mask decoder
- 3D depth-convolution adapter(**DconvAdapter**), 3D depth-MLP adapter(**DMLPAdapter**)
- **SOTA** on AMOS, ACDC, Synapse

@seg_organ — MaskSAM: Auto-prompt SAM [5]



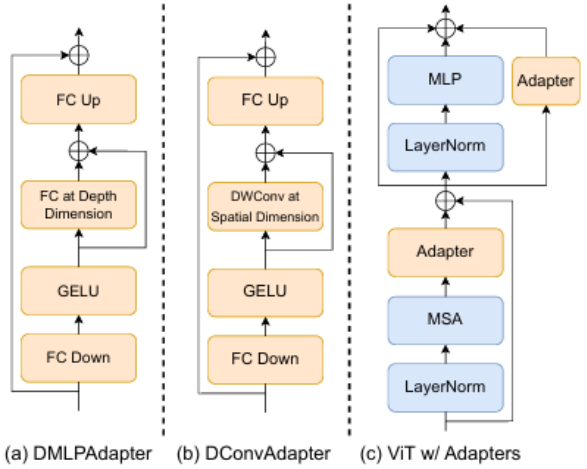


Figure 4. The proposed adapters.

```
if len(xs.shape) == 4:
    xt = rearrange(xs, '(b d) h w c -> b c d h w', d=self.num_frames)
else:
    xt = rearrange(xs, '(b d) (h w) c -> b c d h w', d=self.num_frames,  w=self.size[1])

xt = self.T_DWconv(xt.contiguous())
xt = self.norm(xt)
xt = self.act(xt)
if len(x.shape) == 4:
    xt = rearrange(xt, 'b c d h w -> (b d) h w c')
else:
    xt = rearrange(xt, 'b c d h w -> (b d) (h w) c')

xs = xs + xt
```

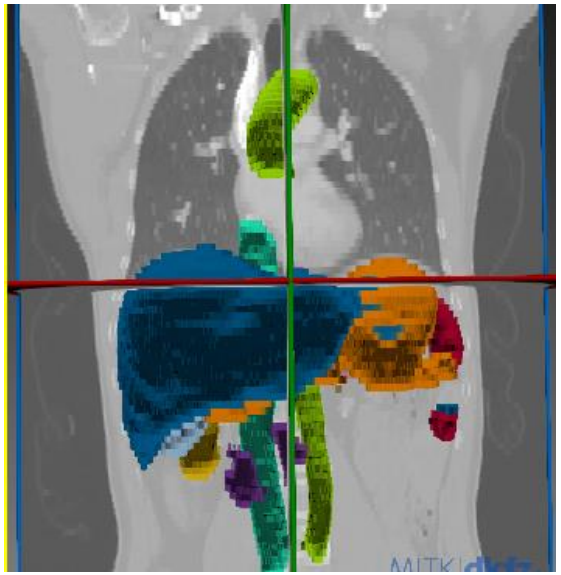
@seg_organ — MaskSAM: Auto-prompt SAM [5]

Paper reported Experiments Result

1000 epochs

Semantic labels	Prompts	Method	Spl.	R.Kd	L.Kd	GB	Eso.	Liver	Stom.	Aorta	IVC	Panc.	RAG	LAG	Duo.	Blad.	Pros.	DSC
✔	–	TransBTS [42]	0.885	0.931	0.916	0.817	0.744	0.969	0.837	0.914	0.855	0.724	0.630	0.566	0.704	0.741	0.650	0.792
		UNETR [17]	0.926	0.936	0.918	0.785	0.702	0.969	0.788	0.893	0.828	0.732	0.717	0.554	0.658	0.683	0.722	0.762
		nnFormer [46]	0.935	0.904	0.887	0.836	0.712	0.964	0.798	0.901	0.821	0.734	0.665	0.587	0.641	0.744	0.714	0.790
		SwinUNETR [16]	0.959	0.960	0.949	**0.894**	0.827	0.979	0.899	0.944	0.899	0.828	0.791	0.745	0.817	0.875	0.841	0.880
		nn-UNet [22]	**0.965**	0.959	0.951	0.889	0.820	0.980	0.890	0.948	0.901	0.821	0.785	0.739	0.806	0.869	0.839	0.878
✗	nnUNet	SAM [26] bbox	0.679	0.741	0.640	0.168	0.443	0.773	0.671	0.651	0.554	0.434	0.232	0.324	0.444	0.698	0.602	0.538
✗	nnUNet	SAM 2 [35] bbox	0.784	0.817	0.819	0.664	0.734	0.780	0.697	0.793	0.739	0.536	0.457	0.604	0.563	0.744	0.691	0.695
✗	nnUNet	MedSAM [31] bbox	0.714	0.811	0.702	0.193	0.469	0.759	0.725	0.701	0.681	0.434	0.365	0.412	0.462	0.783	0.758	0.600
✔	No needs	SAMed [44]	0.849	0.857	0.830	0.573	0.733	0.894	0.816	0.855	0.784	0.727	0.622	0.683	0.701	0.844	0.819	0.772
✔	No needs	SAM3D [5]	0.796	0.863	0.871	0.428	0.711	0.908	0.833	0.878	0.749	0.699	0.564	0.607	0.635	0.884	0.840	0.751
✔	No needs	MaskSAM (Ours)	0.963	**0.973**	**0.969**	0.872	**0.876**	**0.982**	**0.940**	**0.962**	**0.922**	**0.888**	**0.794**	**0.813**	**0.851**	**0.920**	**0.854**	**0.905**

Table 1. The comparison of MaskSAM with SOTA methods on the AMOS testing dataset evaluated by Dice Score. To fair comparison, all results are based on 5-fold cross-validation without any ensembles. "Semantic labels" indicate the model's ability for semantic labeling, while "Prompt" specifies the source of the prompt. The best results are indicated as in **bold**.

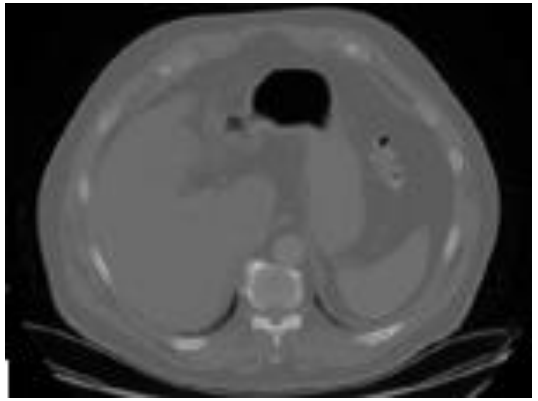


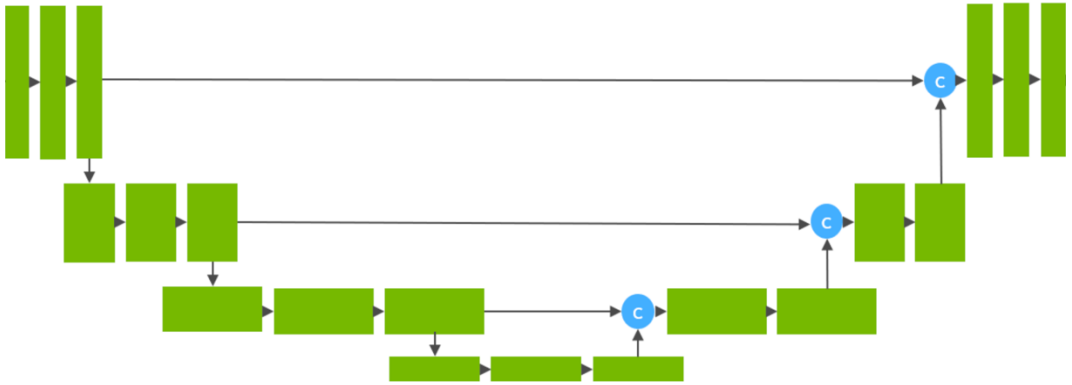
Train Result – 150 epoch

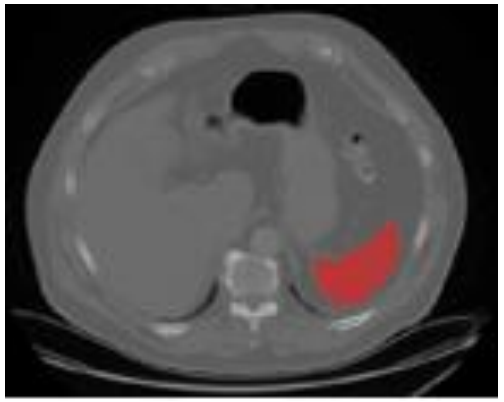
1 epoch: about 8 minutes

	Average	spl.	R.kd	L.kd	GB	Eso	Liver	Stom	Aorta	IVC	Panc	RAG	LAG	Duo	Blad	Pros
Dice	0.8587	0.963	0.949	0.947	0.760	0.823	0.969	0.901	0.945	0.891	0.849	0.706	0.736	0.764	0.866	0.812
IoU	0.7772	0.930	0.909	0.909	0.674	0.706	0.941	0.837	0.898	0.805	0.746	0.556	0.594	0.634	0.795	0.722

@seg_spleen — Spleen Segmentation







Input

- 3D abdominal CT (NIfTI)
- Input Size: 96 x 96 x 96

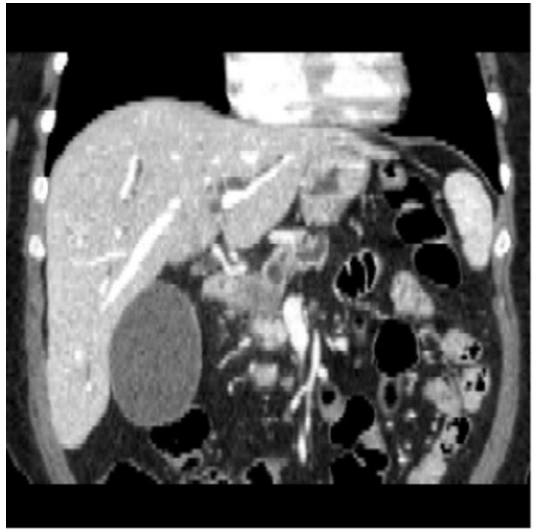
Model

- Residual U-Net [7]
- MONAI bundle: spleen ct segmentation [9]

Output

- Binary spleen mask

@seg_pancreas — Pancreas + Tumor Segmentation



Input

- 3D abdominal CT (NIfTI)
- Input Size: 96 x 96 x 96

Model

- DiNTS (NAS-discovered) [8]
- MONAI bundle: pancreas and tumor DiNTS segmentation [9]

Output

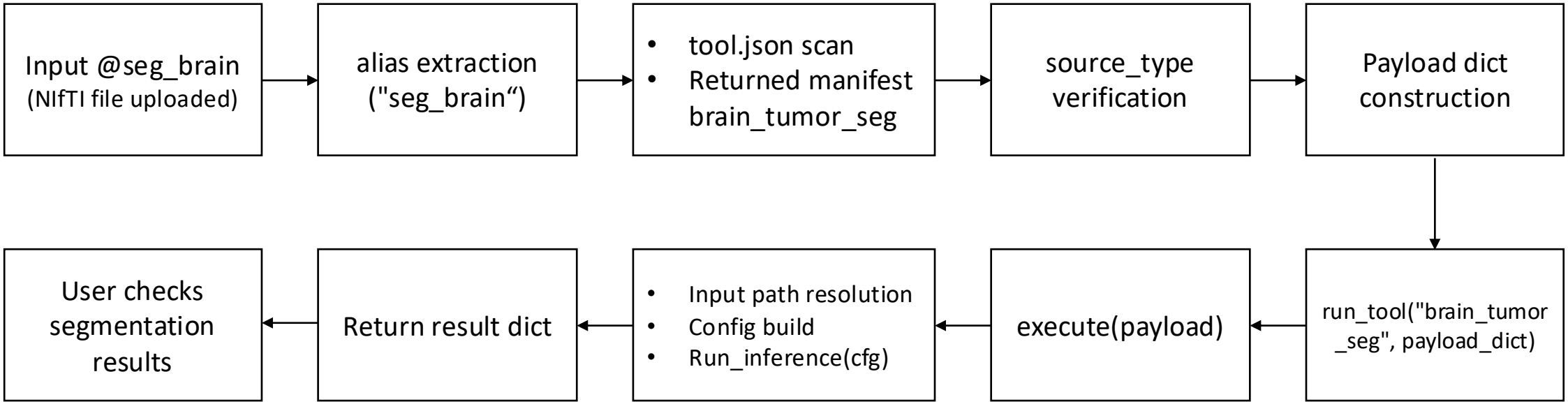
- 3-class mask
- Pancreatic Tumor / Pancreas / Everything else

Skill Patch

Why the orchestrator should call it?

- Alias-based routing with built-in modality guard
  - @seg_* keywords are matched to each tool.json's alias, and source_types auto-blocks invocations whose uploaded source doesn't match — preventing wrong-modality execution
- 1:1 trigger-to-tool mapping: each tool registers a unique alias in trigger_keywords, so routing has no conflicts; the user only picks the alias, the system handles source_type validation
  - Brain tumor MRI: @seg_brain (source_types: ["nifti"])
  - Chest X-ray: @seg_lung (source_types: ["image"])
  - Abdominal CT: @seg_organ / @seg_pancreas / @seg_spleen (source_types: ["nifti"])

Routing


Input @seg_brain
(NIfTI file uploaded)
alias extraction
("seg_brain“)
• tool.json scan
• Returned manifest
brain_tumor_seg
source_type
verification
Payload dict
construction
run_tool("brain_tumor
_seg", payload_dict)
execute(payload)
• Input path resolution
• Config build
• Run_inference(cfg)
Return result dict
User checks
segmentation
results

Routing

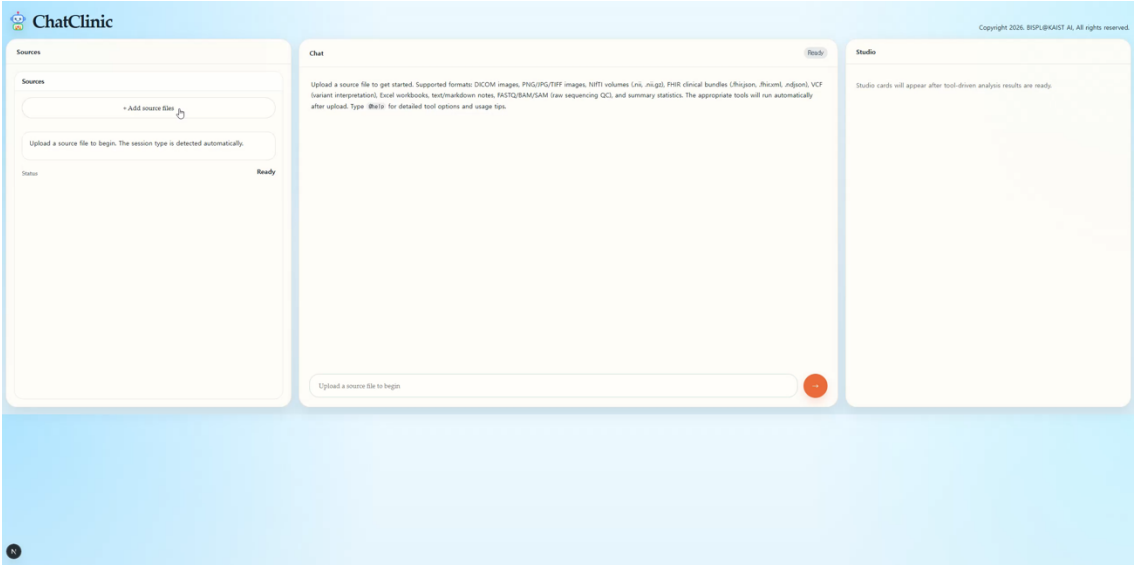
```python
def _run_direct_tool_for_source(
    source_type: str,
    payload: AnalysisChatRequest | RawQcChatRequest | SpreadsheetChatRequest | SummaryStatsChatReque
    tool_request: dict[str, object],
) -> AnalysisChatResponse | RawQcChatResponse | SpreadsheetChatResponse | SummaryStatsChatR  se | Te
    direct_chat = _tool_request_direct_chat_metadata(tool_request)
    endpoint = str(direct_chat.get("endpoint") or "").strip().lower()
    executor = DIRECT_TOOL_ENDPOINT_EXECUTORS.get(source_type, {}).get(endpoint)
    if executor is None:
        return None
    options = _parse_direct_tool_options(
        str(tool_request.get("remainder") or ""),
        str(direct_chat.get("argument_mode") or ""),
    )
    return executor(payload, tool_request, direct_chat, options)
```

```python
DIRECT_TOOL_ENDPOINT_EXECUTORS: dict[str, dict[str, Any]] = {
    "vcf": {
        "liftover": _execute_analysis_direct_liftover,
        "snpeff": _execute_analysis_direct_snpeff,
        "ldblockshow": _execute_analysis_direct_ldblockshow,
        "plink": _execute_analysis_direct_plink,
        "vcf-qc": _execute_analysis_direct_vcf_qc,
        "vcf-interpretation": _execute_analysis_direct_vcf_interpretation,
        "vcf-review": _execute_analysis_direct_vcf_review,
    },
    "raw_qc": {
        "samtools": _execute_raw_qc_direct_samtools,
    },
    "summary_stats": {
        "qqman": _execute_summary_stats_direct_qqman,
    },
    "text": {},
    "spreadsheet": {},
}
```

- Attempting to invoke matched tool executor via DIRECT_TOOL_ENDPOINT_EXECUTORS
  - However, implementation for NIfTI and Image segmentation executors is not yet available. (Future works)

Demo

Demo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Sources
Sources
+ Add source files
Upload a source file to begin. The session type is detected automatically.
Status
Ready
Chat
Ready
Upload a source file to get started. Supported formats: DICOM images, PNG/JPG/TIFF images, NIfTI volumes (.nii, .nii.gz), FHIR clinical bundles (.fhir.json, .fhir.xml, .ndjson), VCF (variant interpretation), Excel workbooks, text/markdown notes, FASTQ/BAM/SAM (raw sequencing QC), and summary statistics. The appropriate tools will run automatically after upload. Type @help for detailed tool options and usage tips.
Upload a source file to begin
Studio
Studio cards will appear after tool-driven analysis results are ready.

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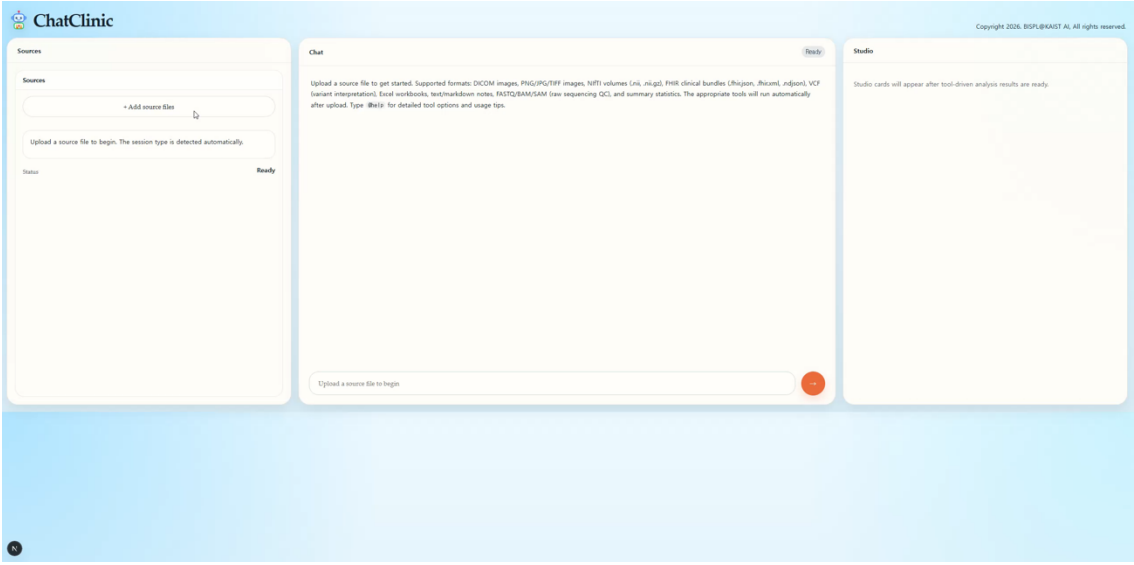
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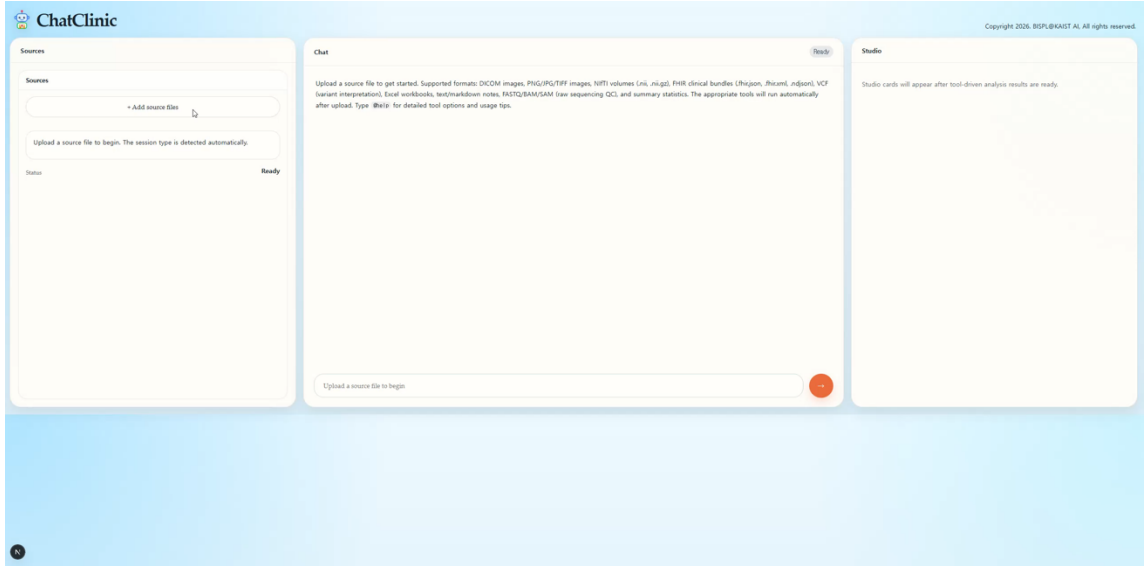
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Sources
Sources
+ Add source files
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Chat
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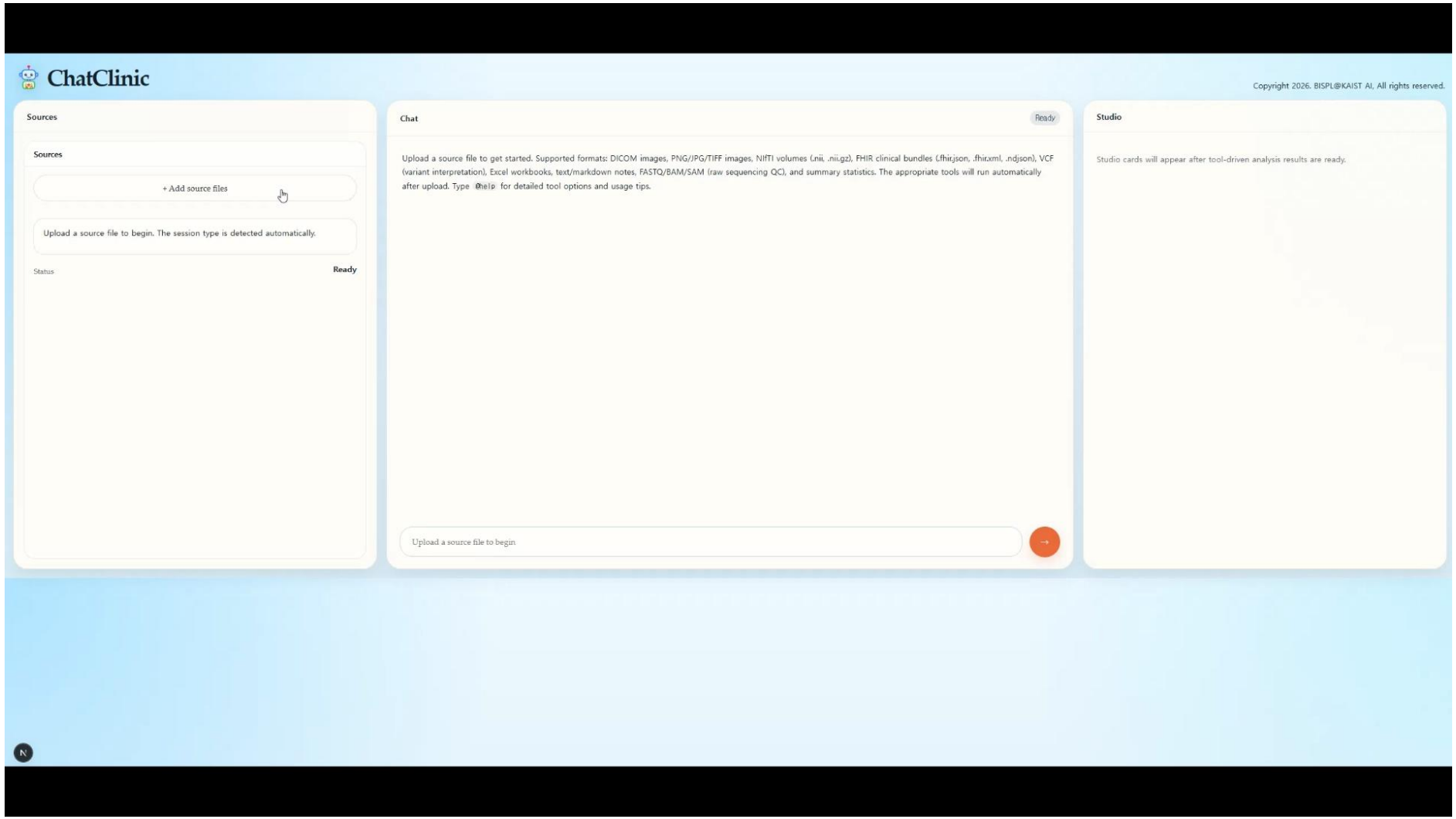
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+ Add source files
Upload a source file to begin. The session type is detected automatically.
Status
Ready
Chat
Ready
Upload a source file to get started. Supported formats: DICOM images, PNG/JPG/TIFF images, NIfTI volumes (.nii, .nii.gz), FHIR clinical bundles (.fhir.json, .fhir.xml, .ndjson), VCF (variant interpretation), Excel workbooks, text/markdown notes, FASTQ/BAM/SAM (raw sequencing QC), and summary statistics. The appropriate tools will run automatically after upload. Type @help for detailed tool options and usage tips.
Upload a source file to begin
Studio
Studio cards will appear after tool-driven analysis results are ready.

Summary

Five tools extend ChatClinic with **medical image segmentation.**

- **Full modality coverage** — MRI, CT, X-ray all supported across 5 tools.
- **@seg_organ trained by us** — MaskSAM + nnU-Net v2 on AMOS22.
- **Format-safe routing** — source_type + @seg_* alias manifest blocks wrong-format invocations.
- **End-to-end integration** — Studio renderer wired into the existing ChatClinic workflow.

Thank You

References

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